Week 6 Progress Report

BSE4104 Emerging Trends in Software Engineering

AI-Native & Agentic Engineering Capstone Project

SME Procurement-Preparation Agent

GROUP J EVENING

1. Work Completed Against Weekly Objectives
    - Designed an explicit Session State model that tracks current item, inventory status, quotations, draft ID, steps taken and agent status during a run.
    - Selected and justified a persistent-memory use case: Preferred Suppliers Memory.
    - Wrote a complete Memory Design and Data Handling Note covering what is stored, why, who can access it, retention and deletion rules.
    - Implemented the preferred suppliers memory in memory.py with functions to add, remove, retrieve and rank quotations using preferences.
    - Integrated memory into the existing get_quotations tool so preferred suppliers are highlighted at the top of the list without taking control of decisions.
    - Demonstrated that memory improves a legitimate task of faster identification of trusted suppliers while remaining strictly advisory.
    - Documented the create_draft_requisition capability as an MCP-style interface, including inputs, outputs, permissions and security boundary.
    - Updated the architecture diagram to show Session State and Persistent Memory layers.
    - Committed all new code, design notes, and specifications to GitHub.

    The agent now maintains explicit state during a run and can remember user-declared supplier preferences across sessions in a safe, auditable way.

2. Key Engineering Decisions and Rationale

Decision	Rationale
Chose Preferred Suppliers as the memory use case	Directly useful for procurement officers, easy to implement, and clearly limited to ranking/suggestion only.
Stored memory in a simple JSON file	Transparent, easy to inspect, sufficient for the project scale, and requires no external database.
Memory only reorders and labels quotations	Guarantees that memory cannot silently select a supplier or bypass the human approval gate.
Documented create_draft_requisition as MCP-style interface	Satisfies the interoperability requirement while keeping the interface tightly scoped and safe.
Explicit retention and deletion rules	Demonstrates responsible data handling even for synthetic operational data.

3. Challenges Encountered and Current Response

Challenge	**Current Status**
Ensuring memory cannot influence critical decisions	Rank and-label only design + clear documentation + code-level separation from approval logic.
Keeping memory simple yet demonstrable	File-based JSON store with a clean API proved sufficient and easy to test.
Making the MCP style specification useful	Focused on one real capability which was create_draft_requisition with full schemas and security boundary.
Avoiding scope creep into full user management.	Limited memory operations to item + supplier preferences only.

4. Repository and Task Management Evidence
GitHub Repository: https://github.com/Nicholas-Muwonge/SME-Procurement-Agent
src/memory.py - Preferred suppliers memory implementation
docs/memory/Memory_Design_and_Data_Handling_Note_v1.0.md
docs/interoperability/MCP_Style_Interface_create_draft_requisition_v1.0.md
Updated src/tools.py
ClickUp Board: https://app.clickup.com/1200420000004171/v/l/f/1200420000011825

5. Plan for Week 7
    - Create a final evaluation set of at least 30 scenarios.
    - Define measurable criteria.
    - Strengthen traces for model, prompt, retrieval, tool, latency and outcome evidence.
    - Implement input-output validation, tool allow-list, authorization, iteration limits, approval controls and prompt injection awareness.
    - Create a Failure Catalogue with at least five genuine failures, remediation and re-test evidence.